



CSE260: Digital Logic Design

Summer 2025

Quiz - 01 [C01]

Total Mark: 15

Duration: 20 Minutes

A

|       |     |          |
|-------|-----|----------|
| Name: | ID: | Section: |
|-------|-----|----------|

1. At BRAC University, professors counted  $(C8)_{16}$  sleepy head-nods on the first day of 8:00 AM classes. Each day after that, head-nods increased by  $(11)_8$  because students kept pulling all-nighters. Professors tracked this for  $(4)_{10}$  days. At the end of day 4, they calculated the grand total head-nods and divided it equally among  $(12)_6$  student councils (keep the divisor in base 6 form, don't convert it to decimal for the final division). [10 Mark]

2. Represent -57 and +49 using 10-bit 2's complement number system. Perform the addition operation on the values found using 10-bit 2's complement. Justify whether there is an overflow or not. [5 Marks]

$$(C8)_{16} = 8 \times 16^0 + C \times 16^1 = 8 + 192 = (200)_{10}$$

$$(11)_8 = 1 \times 8^0 + 1 \times 8^1 = 1 + 8 = (9)_{10}$$

$$\text{Total} = 200 + (200 + 9)$$

$$1^{\text{st}} \text{ day} = 200$$

$$2^{\text{nd}} \text{ day} = 200 + 9 = 209$$

$$3^{\text{rd}} \text{ day} = 209 + 9 = 218$$

$$4^{\text{th}} \text{ day} = 218 + 9 = 227$$

$$\therefore \text{Total} = (854)_{10}$$

|   |     |   |
|---|-----|---|
| 6 | 854 |   |
| 6 | 142 | 2 |
| 6 | 23  | 4 |
| 6 | 3   | 5 |
|   | 0   | 3 |

$(3542)_6$

$$\begin{array}{r}
 12 \mid 3542 \mid 0254 \\
 \hline
 0 \downarrow \\
 35 \\
 \hline
 24 \downarrow \\
 114 \\
 \hline
 104 \downarrow \\
 0402 \\
 \hline
 52 \\
 \hline
 10
 \end{array}$$

$$12 \times 0 = 0$$

$$12 \times 1 = 12$$

$$12 \times 2 = 24$$

$$12 \times 3 = 40$$

$$12 \times 4 = 52$$

$$12 \times 5 = 104$$

2) ~~57~~ unsigned 57 = 111001

$$+ 57 = 0111001$$

$$+ 57 \text{ in 10 bits} = 0000111001$$

$$1111000110$$

$$+ 1$$

$$(-57) \text{ in 10 bits} = (1111000111)_{2's}$$

$$\text{-unsigned } 49 = 110001$$

$$(+49) = 0110001$$

$$(+49) \text{ in 10 bits} = 0000110001$$

$$\therefore +49 = 0000110001$$

$$-57 = 1111000111$$

$$\begin{array}{r}
 0000110001 \\
 + 1111000111 \\
 \hline
 1111110000
 \end{array}$$

As, the operands are in different sign, NO overflow



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B

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1. At BRAC University, professors counted  $(C8)_{16}$  sleepy head-nods on the first day of 8:00 AM classes. Each day after that, head-nods decreased by  $(11)_8$  because students kept pulling all-nighters. Professors tracked this for  $(4)_{10}$  days. At the end of day 4, they calculated the grand total head-nods and divided it equally among  $(12)_6$  student councils (keep the divisor in base 6 form, don't convert it to decimal for the final division). [10 Mark]
2. Represent -49 and +57 using 10-bit 1's complement number system. Perform the addition operation on the values found using 10-bit 1's complement. Justify whether there is an overflow or not. [5 Marks]

Handwritten calculations for Question 1:

$$1) (C8)_{16} = (200)_{10}$$
$$(11)_8 = (9)_{10}$$

1st day = 200  
2nd day = 191  
3rd day = 182  
4th day = 173

$$\text{Total} = (746)_{10}$$

Handwritten calculations for Question 2:

$$(746)_{10} = (3242)_6$$
$$\therefore \frac{(3242)_6}{(12)_6} = (233)_6 \text{ remainder } (2)_6$$

$$\text{+49} = 0000110001$$

$$-49 = (1111001110)_{1's}$$

$$+57 = 0000111001$$

$$\begin{array}{r} \begin{array}{ccccccc} & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 \end{array} & 001 \\ \hline 1 & 1 & 1 & 1 & 001 & 110 \\ \hline 1 & 0000 & 000 & 111 \\ \hline & \xrightarrow{+1} & & \\ \hline 0 & 0000 & 000 & 1000 \end{array}$$

No overflow